

9 Equation Flow Summary

This section presents the complete signal chain as a single, self-contained narrative — from sensor reading to actuator output — showing exactly how each equation feeds into the next at every simulation step.

Step-by-step signal chain

At every simulation step t , the following sequence executes:

(1) Read plant state. The cart-pendulum plant outputs its 8D state:

$$\mathbf{s}_{\text{cart}}(t) = [x, y, \alpha, \beta, \dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}]^\top \in \mathbb{R}^8$$

A demultiplexer splits this into four 2-vectors $[\mathbf{p}, \boldsymbol{\theta}, \dot{\mathbf{p}}, \dot{\boldsymbol{\theta}}]$, and a multiplexer re-assembles the *cart-only* sub-state:

$$\mathbf{s}_c(t) = [x, y, \dot{x}, \dot{y}]^\top \in \mathbb{R}^4$$

(2) Muscle dynamics — low-pass filter on neural command. The LQR command $\mathbf{u} \in \mathbb{R}^2$ (from the previous ZOH slot) drives first-order actuator dynamics:

$$\dot{\mathbf{F}}(t) = \frac{-\mathbf{F}(t) + \mathbf{u}(t)}{\tau_m} \implies \mathbf{F}(t) \in \mathbb{R}^2$$

The muscle state $\mathbf{F}$ is (a) fed to the **ZFT reference mass only** (not to the impedance block directly), and (b) stacked into the 14D state for the LQR. $\mathbf{F}$ reaches the impedance output only indirectly, by shaping how the reference mass moves.

(3) ZFT reference mass — smooth reference trajectory. The virtual reference mass is driven by muscle force and the error to the cart:

$$\ddot{\mathbf{p}}_{\text{ref}}(t) = \frac{1}{M_{\text{ref}}} [\mathbf{F}(t) + K(\mathbf{p}(t) - \mathbf{p}_{\text{ref}}(t)) + D(\dot{\mathbf{p}}(t) - \dot{\mathbf{p}}_{\text{ref}}(t))]$$

Integrating gives:

$$\mathbf{s}_{\text{ref}}(t) = [x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]^\top \in \mathbb{R}^4$$

(4) Impedance force — virtual spring-damper actuation. Using only $\mathbf{s}_c$ (actual cart state) and $\mathbf{s}_{\text{ref}}$ (ZFT output), the impedance block computes the external force applied to the cart:

$$\mathbf{F}_{\text{imp}}(t) = K(\mathbf{p}_{\text{ref}} - \mathbf{p}) + D(\dot{\mathbf{p}}_{\text{ref}} - \dot{\mathbf{p}}) \in \mathbb{R}^2$$

Note: $\mathbf{F}$ (muscle force) is *not* a direct input here. It enters this equation only indirectly through $\mathbf{p}_{\text{ref}}$, which was shaped by the ZFT ODE in Step 3. This is the *sole actuation input* to the cart-pendulum plant. The pendulum angles α, β are shaped indirectly by the resulting cart acceleration.

(5) Full 14D state assembly. All sub-states are concatenated:

$$\mathbf{x}(t) = \underbrace{[x, y, \alpha, \beta]}_{\text{cart pos.}} \oplus \underbrace{[\dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}]}_{\text{cart vel.}} \oplus \underbrace{[F_x, F_y]}_{\text{muscle}} \oplus \underbrace{[x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]}_{\text{ZFT ref.}} \in \mathbb{R}^{14}$$

A Zero-Order Hold (ZOH, $\Delta = 10$ ms) samples this to $\hat{\mathbf{x}}(t)$ before passing it to the LQR.

(6) Finite-horizon LQR — optimal neural command. The state error $e(t) = \hat{x}(t) - x^*$ enters the time-varying gain computed by the backward Riccati pass:

$$u(t) = - \underbrace{R^{-1}B^\top P(t)}_{K(t)} e(t) \in \mathbb{R}^2$$

This $u(t)$ closes the outer loop back to **Step 2** (muscle input) at the next time step.

(7) IK solver — cart position to joint angles. In parallel, the cart position $[x, y]$ is used as the EE target for the manipulator’s inverse kinematics:

$$q_d(t) = f_{\text{IK}}([x(t), y(t)], q_d(t-\Delta t)) \in \mathbb{R}^2$$

Desired joint velocity is approximated numerically: $\dot{q}_d = (q_d(t) - q_d(t-\Delta t))/\Delta t$.

(8) Computed-torque controller — joint torques. The nonlinear manipulator dynamics are cancelled and a PD law is imposed:

$$\tau(t) = M(q)\ddot{q}_d + C(q, \dot{q})\dot{q} + G(q) + K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) \in \mathbb{R}^2$$

τ is applied directly to the manipulator joints, driving the end-effector to follow the cart trajectory.

Compact causal chain

The complete equation flow can be read as a single chain of dependencies:

$$\begin{array}{l} \underbrace{s_{\text{cart}}}_{\text{plant}} \xrightarrow{\text{demux/mux}} s_c \xrightarrow[\text{ZFT}]{F} s_{\text{ref}} \xrightarrow{\text{imp.}} F_{\text{imp}} \longrightarrow \text{cart plant} \\ s_{\text{cart}} \oplus F \oplus s_{\text{ref}} \xrightarrow{\text{mux}} x_{14} \xrightarrow{\text{ZOH}} \hat{x} \xrightarrow{\text{LQR}} u \xrightarrow{\tau_m} F \quad (\text{closes loop}) \\ s_c \xrightarrow{\text{IK}} q_d, \dot{q}_d \xrightarrow{\text{CTC}} \tau \longrightarrow \text{manip. plant} \end{array}$$

Each arrow represents a Drake system connection. The cart-pendulum loop (top two lines) and the manipulator loop (bottom line) run concurrently inside the same Drake **Diagram** at every integration step.